RKG Wallah Academy

Class 7 Mathematics — Complete Notes

NCERT based chapter-wise notes with formulas and solved examples.

Note: Ye NCERT-based summary notes hain. Diagrams NCERT se apni copy mein banao aur exam se pehle apni book se ek baar mila lo.

1. Integers

• Properties for addition: closure, commutative, associative, additive identity 0.

• Multiplication: closure, commutative, associative, multiplicative identity 1, distributive over addition.

• Sign rules: (+)(+) = +; (−)(−) = +; (+)(−) = −. Division follows the same rule. Division by 0 is undefined.

• a ÷ 1 = a; (−a) ÷ (−a) = 1.

2. Fractions and Decimals

• Multiplication of fractions: product of numerators / product of denominators. Reciprocal of a/b = b/a.

• Division of fraction by fraction: multiply by reciprocal of the divisor.

• Decimals: multiply as whole numbers and put decimal point after counting total decimal places. Dividing by 10, 100, 1000 shifts decimal left.

3. Data Handling

• **Mean** = sum of observations / number of observations. **Median**: middle value when arranged in order (average of two middle values if even). **Mode**: most frequent value.

• Range = highest − lowest value.

• **Probability** = favourable outcomes / total outcomes ($0 \le P \le 1$). A coin: P(H) = 1/2; die: P(even) = 3/6 = 1/2.

• Bar graphs, double bar graphs. Chance and probability.

Q. Find mean, median and mode of 5, 7, 3, 7, 8.

Sol. Mean = 30/5 = **6**. Order: 3, 5, 7, 7, 8 → median = **7**. Mode = **7**.

4. Simple Equations

• Equation: variable with equality sign. Solve by transposing terms (change sign) or performing the same operation on both sides.

• Steps for word problems: choose variable, form equation, solve, check.

Q. Solve $3x - 7 = 11$.

Sol. $3x = 18 \rightarrow$ **$x = 6$**.

5. Lines and Angles

• Complementary angles: sum 90°; supplementary: sum 180°. Adjacent angles share a vertex and an arm. Linear pair: adjacent angles on a straight line (sum 180°). Vertically opposite angles are equal.

• Transversal crossing parallel lines: corresponding angles equal, alternate interior angles equal, interior angles on same side sum to 180°.

• Converse: if corresponding or alternate angles are equal then the lines are parallel.

6. The Triangle and Its Properties

• Median joins vertex to midpoint of opposite side; altitude is perpendicular from vertex to opposite side.

• **Exterior angle property:** exterior angle = sum of two interior opposite angles. **Angle sum property:** 180°.

• Triangle inequality: sum of any two sides > third side.

> **Pythagoras theorem (right triangle): (Hypotenuse)² = (Base)² + (Perpendicular)²**

• Pythagorean triplets: (3, 4, 5), (5, 12, 13), (8, 15, 17), (7, 24, 25).

• Equilateral: all sides and angles (60°) equal. Isosceles: two equal sides and angles opposite them equal.

7. Congruence of Triangles

• Two figures are congruent if they have exactly the same shape and size.

• Criteria: **SSS** (three sides), **SAS** (two sides and included angle), **ASA** (two angles and included side), **RHS** (right angle, hypotenuse and one side).

• CPCT: corresponding parts of congruent triangles are equal.

8. Comparing Quantities

Percentage = (Part / Whole) × 100 Profit = SP − CP Loss = CP − SP

Profit % = (Profit / CP) × 100 Loss % = (Loss / CP) × 100 Discount = MP − SP

Simple interest SI = (P × R × T) / 100 Amount A = P + SI

Q. Find SI on ₹5000 at 8% p.a. for 3 years.

Sol. SI = 5000×8×3/100 = **₹1200**; Amount = ₹6200.

9. Rational Numbers

• Rational number = p/q where p, q are integers and q ≠ 0. Examples: 3/4, −5/7, 0.

• Positive and negative rational numbers; equivalent forms; standard form: lowest terms with positive denominator.

• Between two rational numbers there are infinitely many rational numbers.

• Operations: add, subtract (make denominators equal), multiply (num×num/den×den), divide (multiply by reciprocal).

10. Practical Geometry

• Construct a line parallel to a given line through a point outside (using equal alternate angles).

• Triangle construction: SSS, SAS, ASA, RHS.

11. Perimeter and Area

Rectangle: A = l × b, P = 2(l + b) Square: A = a², P = 4a Triangle: A = ½ × base × height

Parallelogram: A = base × height Circle: C = 2πr, A = πr² (π ≈ 22/7 or 3.14)

• Conversions: 1 m² = 10,000 cm²; 1 hectare = 10,000 m².

Q. Find the area of a circle of radius 7 cm.

Sol. A = πr² = (22/7)×7×7 = **154 cm²**.

12. Algebraic Expressions

• Terms, coefficients, like and unlike terms. Monomial (1 term), binomial (2), trinomial (3), polynomial.

• Add/subtract like terms. Multiplying: (a+b)(c+d) = ac + ad + bc + bd.

• Finding the value of an expression by substituting values.

13. Exponents and Powers

$a^m \times a^n = a^{m+n}$ | $a^m \div a^n = a^{m-n}$ | $(a^m)^n = a^{mn}$ | $a^0 = 1$ | $a^{-n} = 1/a^n$

$(a \times b)^m = a^m \times b^m$ | $(a/b)^m = a^m/b^m$

• Standard (scientific) form: write large numbers as $k \times 10^n$ with $1 \le k < 10$, e.g. 4,50,000 = 4.5×10^5.

14. Symmetry

• Line symmetry and rotational symmetry (angle and order of rotation). Regular polygons have as many lines of symmetry as sides.

• Equilateral triangle: 3 lines, rotational order 3; square: 4 lines, order 4; circle: infinite.

15. Visualising Solid Shapes

- Faces, edges, vertices. **Euler's formula:** F + V − E = 2.
- Nets of cube, cuboid, cone, cylinder. Views of 3-D shapes from front, top and side.
- Cube: 6 faces, 8 vertices, 12 edges. Triangular prism: 5 faces, 6 vertices, 9 edges.